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AHEAD Online

Object Oriented Programming Using Java

Objectives

- Polymorphism.
- Types of polymorphism.

Polymorphism

- *Polymorphism* means many forms.
- Allows one interface to be used for a general class of actions.
- Ability to appear in many forms.
- One interface , multiple methods.

Example

- A dog's sense of smell.
- A dog smell's a cat

  Bark and run after the cat.
- A dog smell's food

  It will salivate and run to its bowl.

Example

Consider a stack program

- Can write a program for three type of stack.
- Data stored in each type of stack may vary(integer, floating point, character)
- Stack operations(push, pop) may be the same.
- In non object oriented environment it requires three stack routines.
- In OOPs you can write a single stack program.

Types of Polymorphism

- Static/Compile-Time Polymorphism
- Dynamic/ Run-Time Polymorphism

Static Polymorphism

- The compiler resolves the polymorphism during the compilation of the program. Also, called static binding.
- Achieved by function overloading and method overloading.

Dynamic Polymorphism

- Resolves dynamically at the runtime rather than compile-time .
- Also, known as dynamic binding or Dynamic Method Dispatch.
- Achieved by method overriding.

Takeaways

- What is Polymorphism?
- Examples.
- Types of Polymorphism.

Reference

[1] Naughton P. and Schildt H., “Java2 Complete Reference”, Eighth Edition, Tata McGraw- Hill, 2011.

[2] Ali Bahrami, “Object Oriented Systems Development”, Second Edition, McGraw-Hill, 2008